Advanced Machine Learning, ETH Zurich

Diffusion models

The forward and backward process, training, and conditioning

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Autumn semester 2026

Subject

Which are real? Which are fake?

Entry

1



W. Peebles and S. Xie, arXiv:2212.09748, Fig. 30, cropped. CC BY 4.0.

Subject

Diffusion models in brief

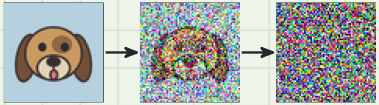
Entry

2

A diffusion model generates new data, such as images, by learning to undo a process that gradually adds noise.

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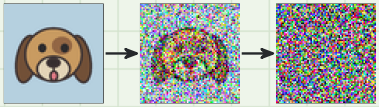
Training



Take real images, add noise in many small steps, and learn to predict the noise that was added.

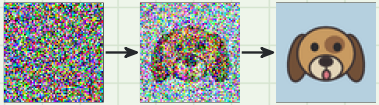
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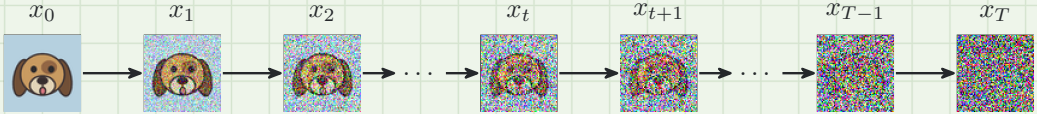
Generation



Start from pure noise and remove the predicted noise, step by step, until a clean image appears.

Each forward step fades the image and adds noise; generation runs the chain backwards.

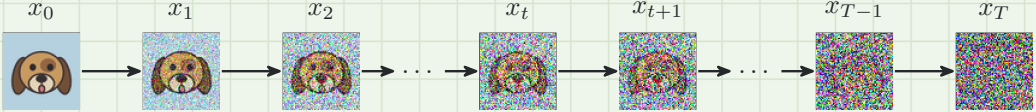
fade & noise



Each forward step fades the image and adds noise; generation runs the chain backwards.

fade & noise

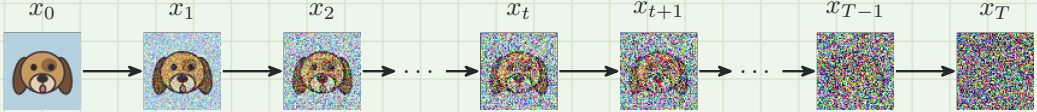
$$\varepsilon_0 + \varepsilon_1 + \varepsilon_2 + \varepsilon_{t-1} + \varepsilon_t = \hat{\varepsilon}_t$$



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fade & noise

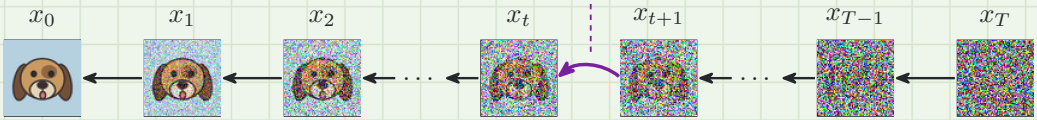
$\varepsilon_0 + \varepsilon_1 + \varepsilon_2 + \varepsilon_{t-1} + \varepsilon_t = \hat{\varepsilon}_t$



If you knew $\hat{\varepsilon}_t$ then you could approximate well x_t from x_{t+1}.

$=$

remove noise



The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t.

1. Take an image of the dataset.

x_0



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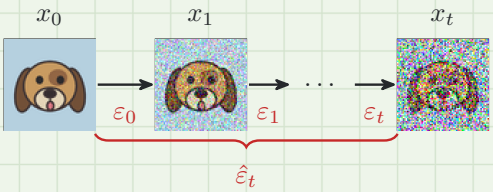
1. Take an image of the dataset.
2. Pick $t \leq T$ at random.

x_0



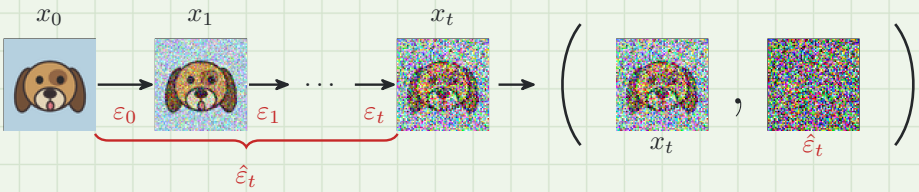
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1. Take an image of the dataset.
2. Pick $t \leq T$ at random.
3. Forward t steps.



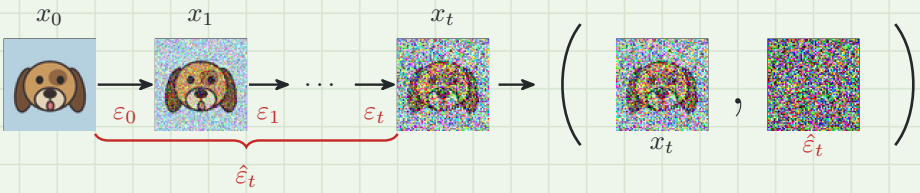
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Repeat until you collect $\{(x_t^{(1)}, \hat{\varepsilon}_t^{(1)}), \ldots, (x_t^{(n)}, \hat{\varepsilon}_t^{(n)})\}$

One neural network, two uses: trained to predict the noise, then used to remove it.

Training



noisy image x_t

model
neural network



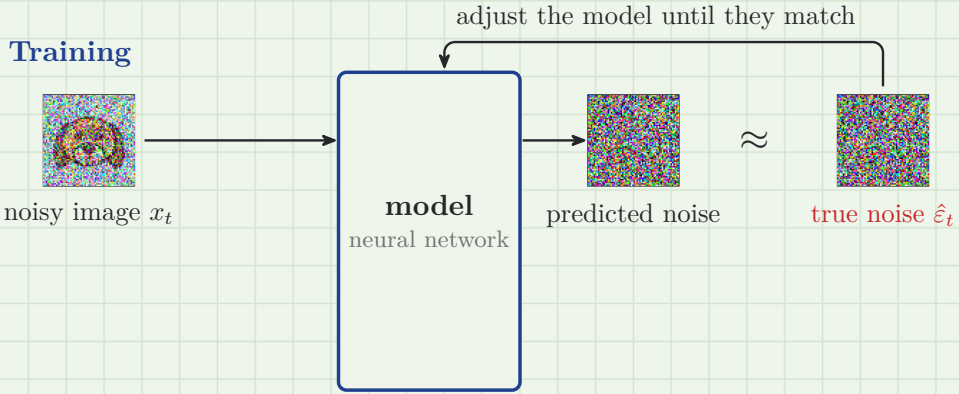
predicted noise

$\approx$

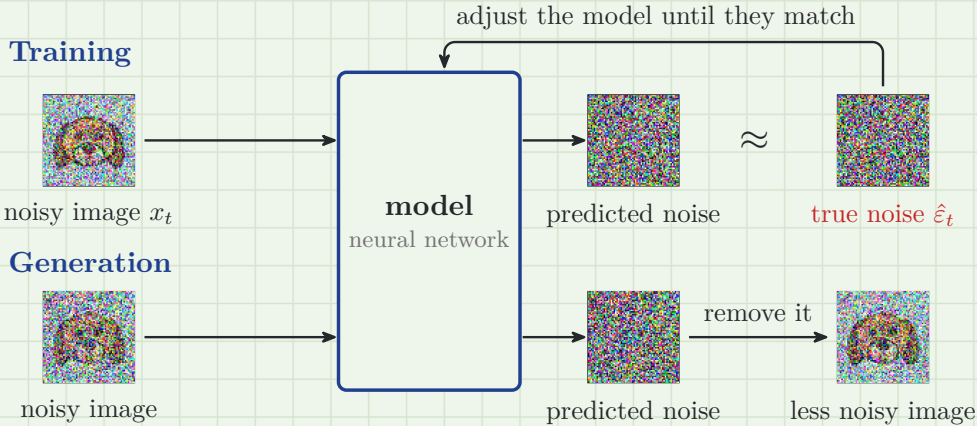


true noise $\hat{\varepsilon}_t$

One neural network, two uses: trained to predict the noise, then used to remove it.



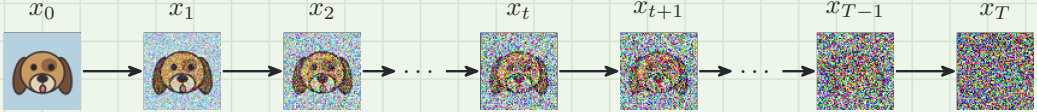
One neural network, two uses: trained to predict the noise, then used to remove it.



The prompt "cat" is given to every backward step, so the image it produces is a cat.

fade & noise

$\varepsilon_0 + \varepsilon_1 + \varepsilon_2 + \varepsilon_{t-1} + \varepsilon_t = \hat{\varepsilon}_t$



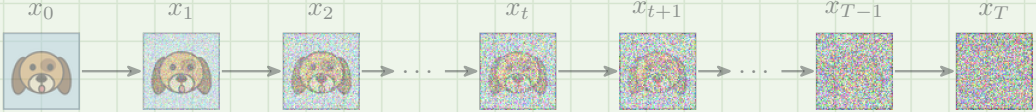
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$=$

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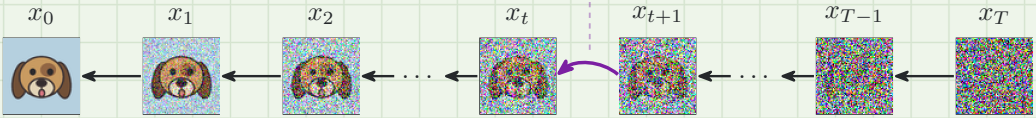
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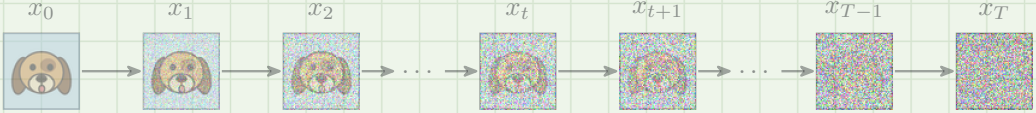
remove noise



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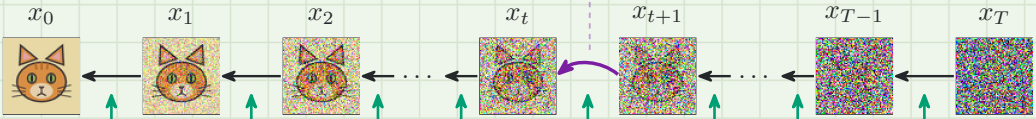
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remove noise



Each training image comes with its caption, which is fed to the model as well.

1. Take an image of the dataset, with its caption.

x_0



"dog"

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2. Pick $t \leq T$ at random.

x_0

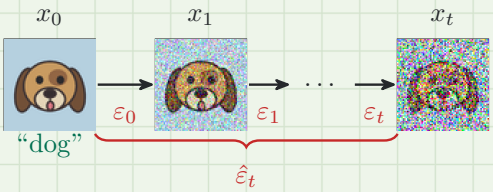


"dog"

Training a conditional model

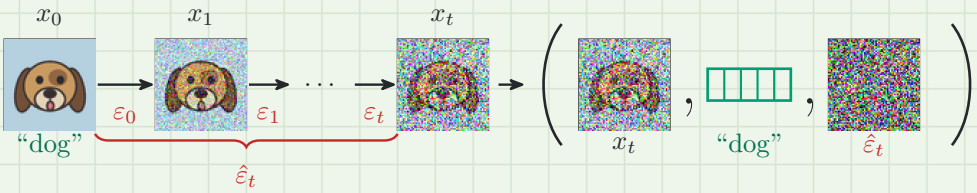
Each training image comes with its caption, which is fed to the model as well.

1. Take an image of the dataset, with its caption.
2. Pick $t \leq T$ at random.
3. Forward t steps.



Each training image comes with its caption, which is fed to the model as well.

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Repeat for many captioned images.